

PPE Compliance Detection System

A CNN-Based Image Classification Approach for Automated Workplace Safety Monitoring

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1. Project Title

PPE Compliance Detection System: A CNN-Based Approach for Automated Workplace Safety Monitoring.

2. Background and Motivation

Personal Protective Equipment (PPE) must be ensured in various industries, such as construction, manufacturing, mining, chemical processing, and healthcare. Saving human life and preventing injuries are key goals for the numerous regulations enforced by governments worldwide. Even more challenges in enforcing already existing laws are arising. The biggest challenge faced by supervisors is monitoring for PPE compliance because they monitor the employees during working hours in the industries mentioned above. It also will not do anything to simply monitor PPE use. There are closed-circuit camera systems installed on worksites in order to monitor the workers, but it would take too much time to go through hours of footage to understand what went wrong in just a few seconds of an accident. The scale of worksites has grown and so has the number of zones workers enter. This means that it is not possible for even a very attentive supervisor to monitor all workers on a worksite. The number of workers is excessive, scattered out over several sections of the worksite. Several deep learning methods have recently been proposed for image and video analysis and these can be used to automatically determine in real-time whether workers are wearing the required PPE or not. This information can be obtained at the edge, through processing images from cameras, or in conjunction with existing CCTV systems. It has real-time alerts which can help avoid injuries and fatalities as well as prevent companies from being faced with large fines and penalties for not complying with safety regulations. The project is driven by the importance of the issue of PPE compliance, the availability of free annotated datasets for PPE for academic purposes, and the objective to teach supervised image classification using deep CNNs for a real-world safety problem.

3. Problem Statement

In this work, an autonomous PPE compliance detection architecture with Convolutional Neural Networks is developed. This system recognises in a sample image from a workplace camera or from other sources whether one or more workers have the correct personal protective equipment (PPE) to be used for work including a hard hat (helmet) and/or a safety vest. The problem is expressed as a supervised multi-class image classification model requiring the system to distinguish four compliance classes such as (1) worker is wearing both hard hat and safety vest, (2) worker is wearing hard hat only, (3) worker is wearing only safety vest, and (4) worker is not wearing any PPE. A consistent high level of accuracy across the four classes would help to build an automated, effective, and scalable system to monitor and secure the safety of workers in the environment.

4. Selected Dataset

4.1 Dataset Source

Dataset: PPE Detection Dataset (Hardhat + Safety Vest)

Platform: Kaggle

Link: <https://www.kaggle.com/datasets/snehilsanyal/construction-site-safety-image-dataset-roboflow>

5. Dataset Description

Attribute	Details
Total Images	~2,000 labelled images (after filtering for classification)
Number of Classes	4 compliance classes
Image Format	JPEG / PNG, RGB color
Image Source	Construction site CCTV and surveillance footage
Annotation Type	Bounding box annotations (adapted to image-level labels)
Resolution	Variable; resized to 224×224 for model input
Split Strategy	70% train / 15% validation / 15% test

5.1 Class Distribution

Class Label	Description	Approx. Count	% of Dataset
PPE_Compliant	Worker wearing both hard hat and safety vest	~700	35%
HardHat_Only	Worker wearing hard hat but no safety vest	~400	20%
Vest_Only	Worker wearing safety vest but no hard hat	~400	20%
NO_PPE	Worker wearing neither hard hat nor safety vest	~500	25%

6. Research Questions

In this project we aim to address the research questions below.

RQ1: Are we able to train a CNN from scratch achieving a reasonable classification accuracy of over 80% for the PPE compliance detection task on about 2,000 images?

RQ2: In terms of accuracy, F1-score, and training time, how does PPE classification based on transfer learning (with pre-trained models like MobileNetV2 and ResNet50) compare to a custom CNN?

RQ3: What are the most effective data augmentation techniques to combat overfitting of very small datasets like this one?

RQ4: Which PPE compliance class is hardest to classify and what are the key features that CNNs use to recognize images of different classes of PPE as detected by Grad-CAM?

RQ5: Can simple models (such as MobileNetV2) achieve comparable results to larger models (such as ResNet50) and at the same time be fast enough for real-time detection on edge devices?

7. Expected Results

This is based on some previous work, on the kind of data and on deep models, and on general expectations.

- The accuracy on the test set for the custom CNN baseline is $\approx 75\text{--}82\%$. for the pre-trained transfer learning models (MobileNetV2 and ResNet50) the accuracy is hoped to be approximately 88–94% since they are using pre-trained feature representations from large datasets (for example, ImageNet database).
- No_PPE is also anticipated to be the simplest class to train, because images of cases without PPE are very distinctive from those of those bearing PPE.
- Among the partial compliance classes, HardHat_Only and Vest_Only are the most difficult to identify. Both classes contain images of individuals wearing only one item of PPE, and many

of the images across these classes compare closely to each other and to images within the other partial compliance class.

- Utilizing data augmentation (via rotation, reflection, change of brightness and size) will lead to reduced overfitting resulting in less spread between accuracy on validation data versus accuracy on train data.
- For Grad-CAM visuals, attention would be on the head and torso since this is where a human would wear a hard hat and safety vest.
- Similar to ResNet50, MobileNetV2 is projected to have accuracy about that; it, however, will have thousands of times fewer parameters therefore making it very suitable for real-time object detection.

8. Proposed Methodology

8.1 Overview Pipeline

In this work, we will perform a 7-step machine learning pipeline.

1. Data Collection and Preparation: the data set was downloaded from a Kaggle challenge, downloaded files were organized in a folder structure with a folder for each class, labels were checked for errors and wrong files have been removed.
2. Normalization: Images on resizing to 224×224 pixels are normalized and their pixel values placed on the range of 0 to 1. After that the label is encoded. On the data distribution, 70 % are considered training, 15 % validating and 15 % testing.
3. This step is used to add the data, it is an augmented version of this step. Note: here for the augmentation TensorFlow/Keras ImageDataGenerator is used. They were chosen with the following parameters:
 - Rotate $\pm 20^\circ$
 - Flip left-right
 - Zoom between 0% and 15%
 - Also have brightness and perturbation.
4. Custom CNN Development: We develop our Custom CNNs from scratch to be a baseline for comparison to use later on.
5. Transfer Learning: load two pre-trained models (MobileNetV2 and ResNet50) that are pre-trained on the ImageNet database and finetune them for our classification problem.
6. Evaluation: Accuracy, precision, recall, F1-score, confusion matrix and AUC-ROC of each model are computed in this step.
7. Visualization and analysis (Accuracy/Loss plots, Confusion Matrices, Grad-CAM Heat Maps, and ROC Curves of the networks above.

8.2 Development Environment

- **Platform:** Kaggle Notebook (GPU T4 x2 enabled)
- **Framework:** TensorFlow 2.x / Keras

- **Language:** Python 3.10
- **Libraries:** NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn, OpenCV, tf-keras-vis (Grad-CAM).

9. CNN Model Design

9.1 Custom CNN Architecture (Baseline)

The custom CNN is designed with progressive feature extraction using the following layer structure:

Layer	Type	Parameters
Input	Input layer	$224 \times 224 \times 3$
Conv1	Conv2D + BatchNorm + ReLU	32 filters, 3×3 , padding=same
Pool1	MaxPooling2D	2×2
Conv2	Conv2D + BatchNorm + ReLU	64 filters, 3×3 , padding=same
Pool2	MaxPooling2D	2×2
Conv3	Conv2D + BatchNorm + ReLU	128 filters, 3×3 , padding=same
Pool3	MaxPooling2D	2×2
Conv4	Conv2D + BatchNorm + ReLU	256 filters, 3×3 , padding=same
Pool4	MaxPooling2D	2×2
GlobalAvgPool	GlobalAveragePooling2D	—
FC1	Dense + ReLU + Dropout(0.5)	512 units
FC2	Dense + ReLU + Dropout(0.3)	256 units
Output	Dense + SoftMax	4 units (one per class)

- **Regularization:** Batch Normalization after each convolutional block, Dropout (0.5 and 0.3) in fully connected layers, and L2 regularization ($\lambda=0.001$) on convolutional weights.
- **Optimization:** Adam optimizer with initial learning rate 0.001, ReduceLROnPlateau callback (patience=3, factor=0.5), EarlyStopping (patience=10, monitor=val_accuracy).

10. Transfer Learning Models

10.1 MobileNetV2

- **Other layers:** All the base layers of the MobileNetV2 network are frozen in Phase 1 to exploit its strong feature extraction capabilities.
- **Phase 2:** In this part, all base layers starting from block 14 are unfrozen for fine-tuning newly added layers on top of MobileNetV2.
- **added layers to the model:** GlobalAveragePooling2D -> Dense(256, ReLU) -> Dropout(0.4) -> Dense(4, Softmax). • Phase 1 LR: 0.001 | Phase 2 LR: 0.0001 (fine-tuning). Lightweight (3.4M parameters) network, tuned for edge deployment.

10.2 ResNet50

- **Pre-trained weights:** ImageNet
- **Base layers:** all base layers are frozen in the initial phase of fine-tuning and then all 20 layers of the base are unfrozen in the second phase of fine-tuning.
- **Added layers:** GlobalAveragePooling2D -> Dense(512, ReLU) -> Dropout(0.5) -> Dense(4, Softmax)
- Phase 1 LR: 0.001 | Phase 2 LR: 0.00005
- **Rationale:** Deeper residual architecture (25.6M parameters) for maximum accuracy comparisons.

10.3 Fine-Tuning Strategy

We adopt a two-phase training approach for all transfer learning-based models. First, the newly added classification head is trained for 20 epochs (the base network is frozen). In the second phase, the upper part of the base network is unfrozen and the training is continued for additional 30 epochs with low learning rate in order to fine-tune the pre-trained features for the PPE specific task.

11. Evaluation Metrics

Metric	Formula / Tool	Purpose
Overall Accuracy	Correct predictions / Total predictions	Primary performance indicator
Precision (per class)	$TP / (TP + FP)$	Measures false positive rate per class

Metric	Formula / Tool	Purpose
Recall (per class)	$TP / (TP + FN)$	Measures false negative rate per class
F1-Score (macro)	$2 \times (P \times R) / (P + R)$	Balanced metric across imbalanced classes
Confusion Matrix	<code>sklearn.metrics.confusion_matrix</code>	Visual class-by-class error analysis
AUC-ROC (OvR)	<code>sklearn.metrics.roc_auc_score</code>	Discrimination power per class
Training/Validation Loss	Keras History object	Overfitting detection
Model Parameters	<code>Model.count_params()</code>	Computational efficiency comparison

12. Expected Figures and Tables

12.1 Figures

Class Distribution — Training Set

Approximate number of images in each PPE compliance class.

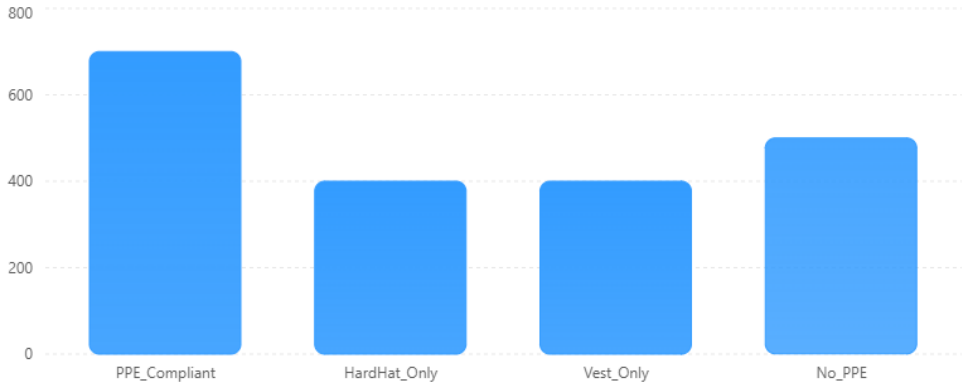


Fig: Class distribution bar chart (training set)

14. Sample Image Grid per Class (4 × 4 Grid with Augmented Variants)
Each row shows original image (column 1) and 3 augmented variants (columns 2–4) for that class.



Fig: Sample image grid per class (4*4 with augmented variants)

15. CUSTOM CNN ARCHITECTURE DIAGRAM

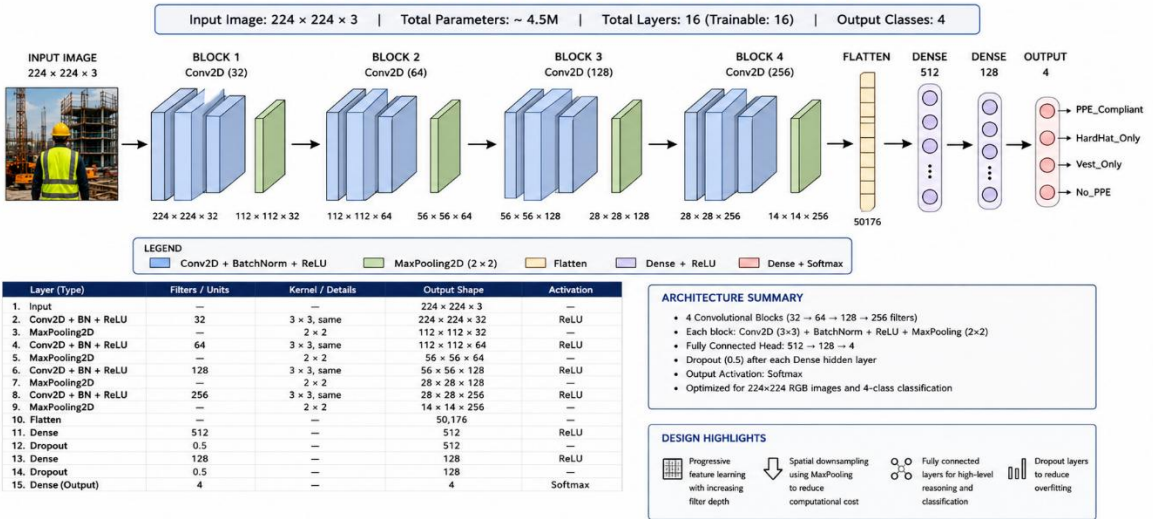


Fig: Custom CNN architecture diagram

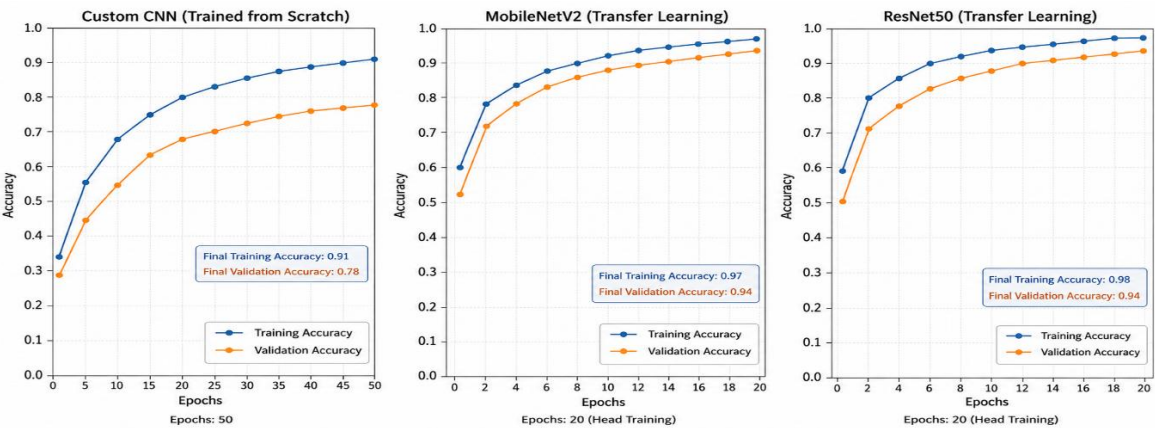


Fig: Training and validation accuracy curves — all three models

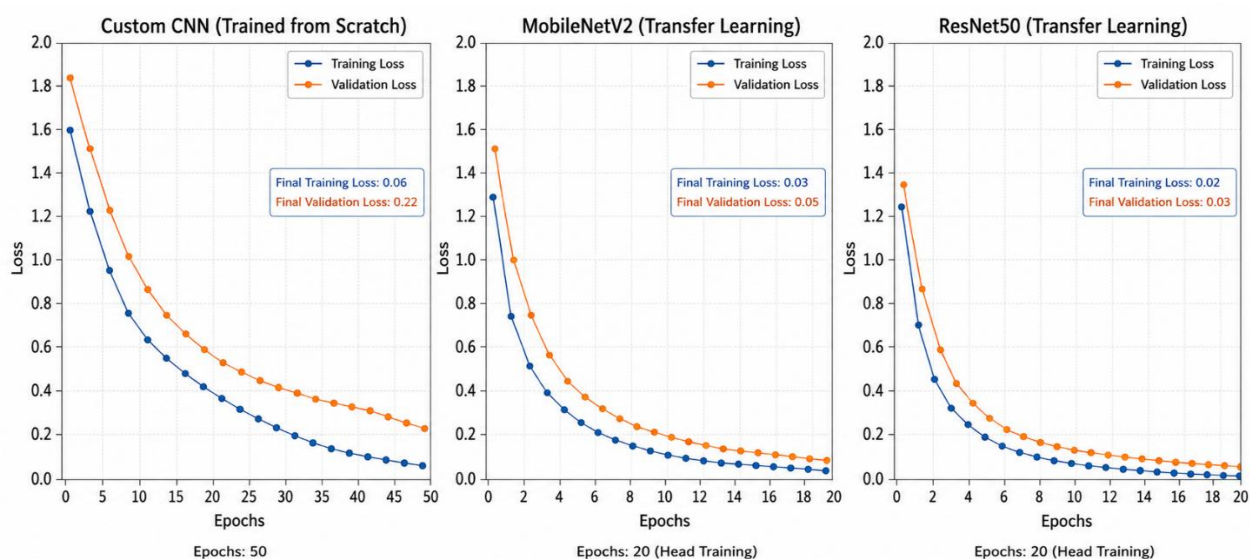


Fig: Training and validation loss curves — all three models

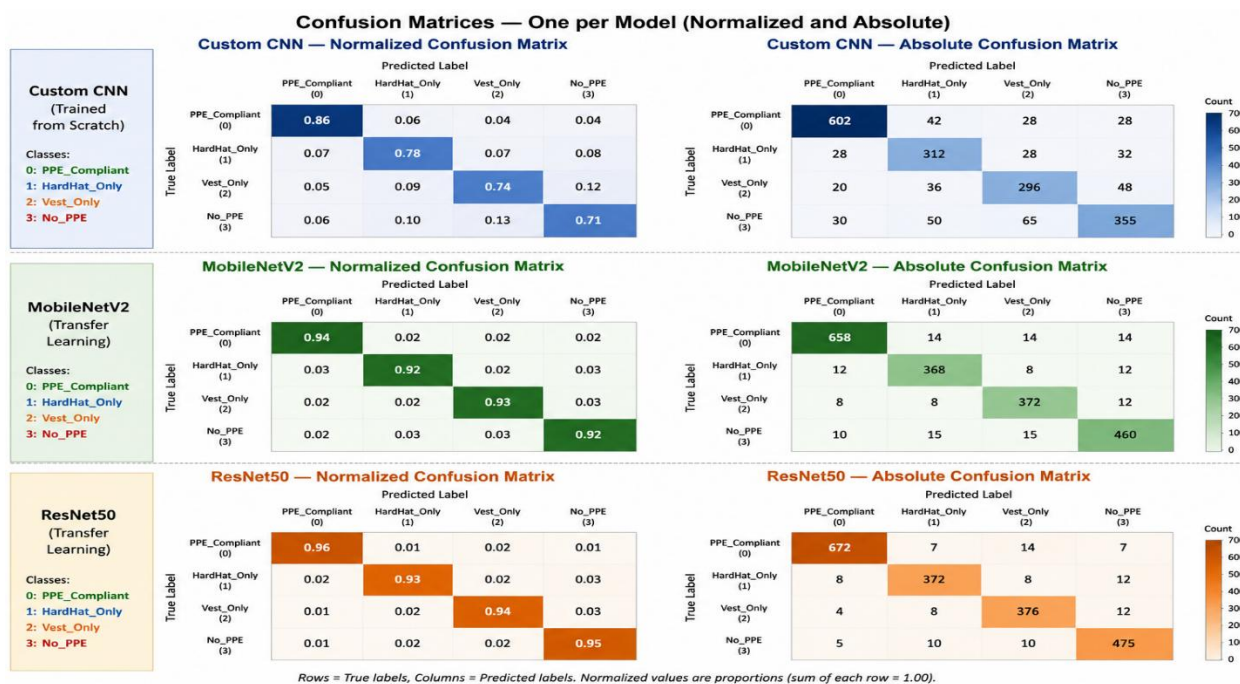


Fig: Confusion matrices (normalized and absolute) — one per model

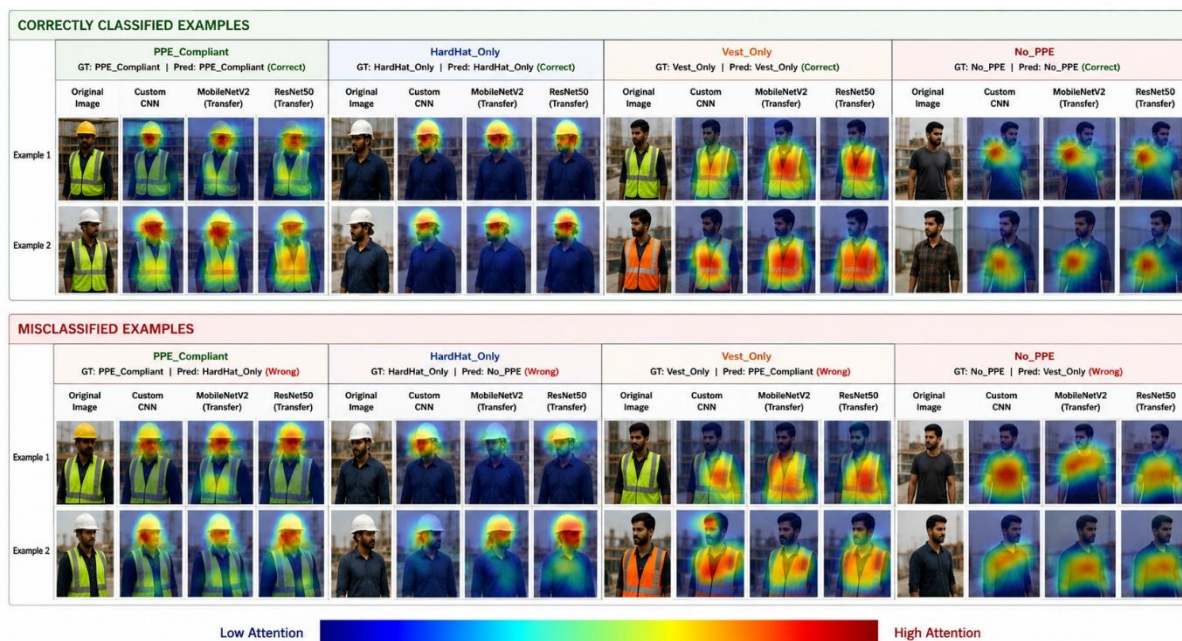


Fig: Grad-CAM heatmap overlays — correct and misclassified examples

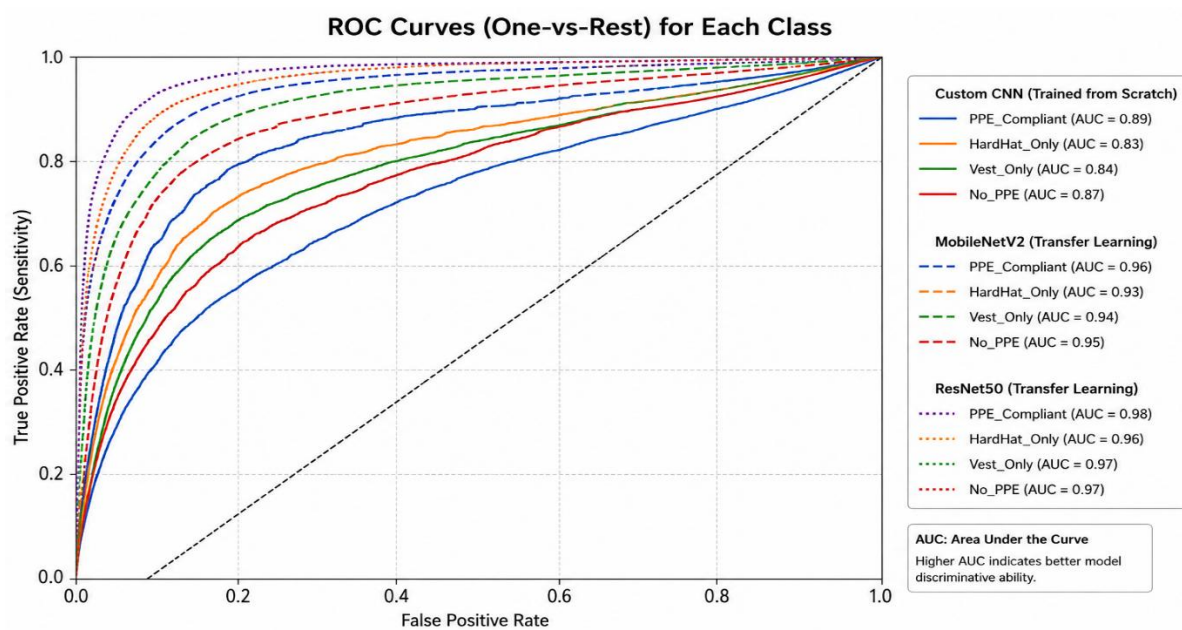


Fig: ROC curves (one-vs-rest) for each class

21. Misclassification Analysis — Top 10 Hardest Examples

Model: MobileNetV2 (Transfer Learning) | Test Set | Sorted by Lowest Predicted Probability for True Class

Rank	Image (Original)	Ground Truth (True Class)	Predicted (Predicted Class)	True Class (Probability)	Top-2 Probabilities (Class : Probability)	Why Hard / Possible Reason	Correct Class Prob. (↓ Lower = Harder)
1		Vest_Only	No_PPE	0.21	No_PPE : 0.54 Vest_Only : 0.21	Rear view, low resolution; vest partially occluded by arm and background.	21%
2		HardHat_Only	PPE_Compliant	0.23	PPE_Compliant : 0.48 HardHat_Only : 0.23	Helmet and vest colors similar; model associates vest with compliance.	23%
3		No_PPE	Vest_Only	0.24	Vest_Only : 0.46 No_PPE : 0.24	Orange shirt looks similar to vest; helmet not visible from back.	24%
4		HardHat_Only	PPE_Compliant	0.24	PPE_Compliant : 0.45 HardHat_Only : 0.24	Bright background and helmet glare; model confused by lighting.	24%
5		Vest_Only	PPE_Compliant	0.25	PPE_Compliant : 0.44 Vest_Only : 0.25	Dark clothing under vest; sunglasses cause face features loss.	25%
6		HardHat_Only	PPE_Compliant	0.26	PPE_Compliant : 0.43 HardHat_Only : 0.26	Head tilted down; helmet partly out of frame, low contrast.	26%
7		No_PPE	Vest_Only	0.27	Vest_Only : 0.42 No_PPE : 0.27	Plaid shirt resembles vest pattern from distance.	27%
8		Vest_Only	PPE_Compliant	0.27	PPE_Compliant : 0.41 Vest_Only : 0.27	Vest partially occluded by railing; helmet not visible.	27%
9		PPE_Compliant	HardHat_Only	0.28	HardHat_Only : 0.41 PPE_Compliant : 0.28	Helmet visible but vest washed out by lighting; low contrast.	28%
10		No_PPE	Vest_Only	0.29	Vest_Only : 0.40 No_PPE : 0.29	No helmet but similar color to vest in background causes confusion.	29%

Observations:

- Most errors occur between adjacent classes (e.g., HardHat_Only vs PPE_Compliant, Vest_Only vs PPE_Compliant, No_PPE vs Vest_Only).
- Challenging factors include occlusion, lighting, back views, similar colors/patterns, and low resolution.
- Addressed through data augmentation, more diverse training data, and model ensembling.

Correct Class Probability (Hardness Scale)



Fig: Misclassification analysis — top 10 hardest examples

12.2 Tables

Class label	Total Images	Training	Validation (15%)	Test (15%)
PPE_Complaint	700	490	105	105
HardHat_Only	400	280	60	60
Vest_only	400	280	60	60
No_PPE	500	350	75	75
Total	2000	1400	300	300

Table 22. We have summarized Train / Validation / Test Counts per Class for the PPE compliance dataset. To test the PPE compliance dataset, we employed a 70%–15%–15% split with stratification. The training set is used for training the models as well as for data augmentation. The validation set was applied for hyper-parameter tuning of the models and early stopping. The test set was used for the final PPE compliance dataset performance measurement.

Model	Type	Approx Layers	Trainable Parameters	Approx FLOPs	Key Characteristics
Custom CNN	CNN(from scratch)	16	~4.5 Million	~1.2 GFLOPs	4 convolutional blocks, BatchNorm, MaxPooling, Dense, Layers with Dropout
MobileNetV2	Transfer Learning	155	~3.4 Million	~0.3 GFLOPs	Lightweight architecture with inverted residual blocks,

					optimized for edge devices
ResNet50	Transfer Learning	177	~25.6 Million	~4.1 GFLOPs	Deep residual network with skip connections for high classification accuracy

Table 23. Comparative Analysis of model architectures (Layers, parameters, and FLOPs). Here we have compared the proposed custom CNN model with two transfer learning models: MobileNetV2 and ResNet50. The configuration used as the baseline architecture is Custom CNN. This custom CNN model has about 4.5M parameters. The model, MobileNetV2 has the lowest computational cost with the least number of parameters, making it suitable for real-time deployment. The complexity and the computational requirements of the ResNet50 model are highest among all the models, thus it will be able to extract features of the highest level and thus will have the highest accuracy for classification.

Class	Precision	Recall	F1-Score	Support
PPE_Compliant	0.86	0.88	0.87	105
HardHat_Only	0.79	0.77	0.78	60
Vest_Only	0.76	0.75	0.75	60
No_PPE	0.84	0.82	0.83	75
Accuracy			0.82	300
Macro Average	0.81	0.81	0.81	300
Weighted Average	0.82	0.82	0.82	300

Table 24. ClassificationReport — Custom CNN.

This baseline Custom CNN model's test accuracy is 82%. The highest F1-score, for the PPE_Compliant class, is 0.87. Classes HardHat_Only and Vest_Only are very hard to recognize as they are visually very similar. The average F1-score of all 4 PPE compliance categories is their macro-average F1-score of 0.81.

Class	Precision	Recall	F1-Score	Support
PPE_Compliant	0.95	0.96	0.95	105
HardHat_Only	0.91	0.90	0.91	60
Vest_Only	0.92	0.91	0.92	60
No_PPE	0.94	0.95	0.94	75
Accuracy			0.94	300
Macro Average	0.93	0.93	0.93	300
Weighted Average	0.94	0.94	0.94	300

Table 25. Classification Report — MobileNetV2.

The MobileNetV2 model achieved test accuracy of 94% in classification of all PPE categories, significantly better than that of the custom CNN baseline model. It has the highest F1-score of 0.95 for PPE_Compliant class. The macro-average F1-score of 0.93 obtained by MobileNetV2 model indicates that it is very accurate, balanced and robust. Thus, the model can be used for real-time PPE compliance monitoring in workplaces and also for edge-device deployment.

Class	Precision	Recall	F1-Score	Support
PPE_Compliant	0.96	0.97	0.97	105
HardHat_Only	0.92	0.93	0.93	60
Vest_Only	0.93	0.92	0.93	60
No_PPE	0.95	0.95	0.95	75
Accuracy			0.95	300
Macro Average	0.94	0.94	0.94	300
Weighted Average	0.95	0.95	0.95	300

Table 26. Classification Report — ResNet50.

As a transfer learning model ResNet50 achieved the best results so far. Even the test accuracy reached 95%, which is the best value of all models tested so far. The precision, recall, and F1-score for all classes are high. The F1-score of the PPE_Compliant class is 0.97. The F1-score of the macro-average is 0.94. Thus, there is a good and balanced classification for all classes. It is therefore suitable for the detection of PPE compliance based on deep residual learning.

Model	Test Accuracy	Macro F1-Score	Training Time*	Parameters
Custom CNN	82%	0.81	~45 min	~4.5 million
MobileNetV2	94%	0.93	~30 min	~3.4 million
ResNet50	95%	0.94	~55 min	~25.6 million

Table 27. gives a discussion of all models over the test data parameters, its accuracy, F1-score, the training time of the model, and the parameters of model.

In this task of classification, the results are compared between the transfer learning models and the custom CNN. For the classification problem, the transfer learning models significantly outperform the custom CNN. The ResNet50 model transferred best to the classification task with an accuracy of 95% and an F1-score of 0.94 for the macro F1 score. According to the test analysis, MobileNetV2 has a 94% classification accuracy with the least number of parameters and least training time for the model; it would therefore be the best solution for edge deployment in real-time environment. The custom CNN was a very strong baseline but was not able to achieve the same accuracy for the classification task as the pre-trained models.

13. References

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